RV College of Engineering®

QB_E&IPR_ Unit II_ 112025

S.No	Question	Marks	BT	CO
1	Topic ID :1 What is a key step in evaluating a business opportunity? ANS: A market analysis and competitor analysis are crucial for understanding the viability of an idea.	2	2	2
2	Topic ID :1 Evaluating an opportunity primarily involves assessing: ANS: Evaluating an opportunity primarily involves assessing the market, the financial viability, the team's ability to execute, and the strategic fit with your goals. This can be broken down into several key areas.	2	2	2
3	Topic ID :2 In the context of ideation, integrating engineering principles primarily serves to---------- ANS: Ensure that generated ideas are technically feasible and practical	2	3	3
4	Topic ID:3 Primary benefit of cross-disciplinary collaboration for technological innovation is---------- ANS: Encourages fresh perspectives and creative problem-solving by combining diverse knowledge and skill sets.	2	3	3

5	Topic ID:4 In demand analysis, what does "ceteris paribus" mean? ANS: Ceteris paribus is a Latin phrase meaning "all other things being equal." In demand analysis, it means that when studying the effect of a price change on quantity demanded, all other factors like consumer income, tastes, and the prices of related goods are assumed to remain constant.	2	2	2
6	Topic ID:5 What is the primary purpose of a prototype in the development process? ANS: To get early user feedback and validate requirements	2	2	2
7	Topic ID: 6 How does a Proof of Concept differ from a prototype? ANS: A PoC proves technical feasibility; a prototype demonstrates the product's functionality and user experience.he core difference is that a PoC answers the question "Can we do this?", while a prototype answers "How will the user interact with this?	2	2	2
8	Topic ID: 7 If total revenue increases by $100 when output changes by one unit, what is the marginal revenue? ANS: Marginal revenue is the change in tiotal revenue from selling one additional uint of output. In the given case, since the total revenue increased by $100 from one more unit, the marginal revenue is simply that change. Or $\frac{\$100}{1} = \$100$	2	3	3

9	Topic ID: 8 What is deferred revenue expenditure?. Give one example. ANS: Deferred revenue expenditure is a revenue expense that is spread over multiple accounting periods because its benefits extend beyond one year. Instead of expensing the entire cost in the current period, a portion is recorded as an asset on the balance sheet and expensed over time, typically a few years. A common example is heavy advertising for a new product launch, where the expense is amortized over the years the campaign is expected to generate benefits.	2	2	2
10	Topic ID:9 A company has fixed costs of $60,000, a variable cost of $8 per unit, and a selling price of $20 per unit. What is the break-even point in units? ANS: Contribution margin per unit = Selling price - Variable cost = $20 - $8 = $12.Break-even point (units) = Fixed Costs / Contribution Margin per Unit = $60,000 / $12 = 5,000 units	2	3	3
	Part B			
1	**Define Market Opportunity and explain the key factors to identify potential market opportunities.** Ans : Market opportunity refers to a situation in which a product or service can fulfill an unmet customer need. Key factors: - Customer pain points and needs - Market size and growth rate - Competitive landscape - Emerging technologies and trends - Regulatory or demographic changes - Availability of resources and infrastructure.	8	2	2

2	**Explain how engineering principles are integrated into the ideation process of entrepreneurship.** Engineering principles such as design thinking, optimization, prototyping, and systems approach help transform ideas into viable technical solutions. Integration involves: - Applying problem-solving methods to real-world challenges - Using data-driven design and simulation tools - Considering sustainability, efficiency, and scalability in ideation - Example: Using CAD tools to develop prototypes for innovative machinery.	8	2	2
3	**Apply Porter's Generic Strategies to suggest how a startup in renewable energy can gain a competitive advantage.** **Solution:** Porter's generic strategies include cost leadership, differentiation, and focus strategies for achieving competitive advantage. **1. Cost Leadership:** - Reduce manufacturing costs through automation and local sourcing. - Example: Large-scale production of low-cost solar panels. **2. Differentiation:** - Offer innovative features such as smart solar panels with IoT integration. - Build brand loyalty through superior quality and after-sales service. **3. Focus Strategy:** - Target niche segments like rural electrification or SMEs. - Customize products to suit specific energy needs. **Conclusion:** A renewable energy startup can use a hybrid approach combining differentiation and cost efficiency for long-term growth.	8	3	3

4	**Analyze how cross-disciplinary collaboration enhances technological innovation in startups.** ** Solution:** Cross-disciplinary collaboration integrates knowledge from different fields to create superior innovations. **Key Aspects:** 1. **Knowledge Integration:** Engineers, designers, and business analysts share expertise. 2. **Innovation Acceleration:** Diverse perspectives lead to creative problem-8solving. 3. **Improved Prototypes:** Collaboration between R&D and marketing ensures user-centered design. 4. **Example:** A biomedical startup combining doctors (medical expertise) and engineers (device design) creates affordable diagnostic devices	8	2	2
5	**Examine the components of market feasibility and explain how demand analysis helps in business decision-making.** ** Solution:** Market feasibility assesses the potential success of a new product or service in a target market. **Components of Market Feasibility:** 1. **Target Market Identification:** Define demographic and geographic scope. 2. **Customer Segmentation:** Categorize based on behavior, needs, or income. 3. **Competitor Analysis:** Study competitors' products and pricing. 4. **Market Trends:** Evaluate growth potential and entry barriers. 5. **Pricing Strategy:** Determine willingness to pay. **Demand Analysis Role:** - Quantifies potential sales volume. - Guides inventory, production, and pricing. - Reduces risk of overproduction.	8	2	2

6	**Evaluate the importance of technical feasibility through prototype development and proof of concept.** **Solution:** Technical feasibility ensures that a business idea is technically possible and implementable. **1. Prototype Development:** - Builds a working model to test design and functionality. - Identifies flaws early, saving costs. **2. Proof of Concept (PoC):** - Demonstrates technical viability of core idea. - Provides confidence to investors. **Evaluation Aspects:** - Performance metrics (speed, accuracy, durability) - Resource requirements and scalability	8	3	3
7	**Assess how financial feasibility analysis influences entrepreneurial decision-making.** **Solution:** Financial feasibility assesses whether the business can generate adequate returns. **Steps in Financial Feasibility:** 1. **Cost Estimation:** Identify fixed and variable costs (machinery, labor, materials). 2. **Revenue Projection:** Forecast income from expected sales. 3. **Break-Even Analysis:** Find point where total cost = total revenue. **Formula:** Break-Even Point = Fixed Costs / (Selling Price – Variable Cost) **Example:** If fixed cost = ₹1,00,000, selling price = ₹200, variable cost = ₹100 ⇒ BEP = 1000 units.	8	2	2
8	**Develop a brief business plan structure for a startup focused on eco-friendly construction materials.** ** Solution:** **1. Executive Summary:** Vision of sustainable construction. **2. Company Description:** Mission to reduce carbon footprint. **3. Market Analysis:** High demand for green building projects. **4. Organization Structure:** Roles – CEO, R&D, Marketing, Operations. **5. Product Line:** Eco-bricks, recycled cement blocks. **6. Marketing Plan:** Target construction firms and government projects. **7. Financial Plan:** Estimate costs, revenues, and profits. **8. Break-even Analysis:** Determine financial sustainability.	8	2	2

9	**Apply SWOC analysis for a drone-based agricultural service startup**. ** Solution:** **Strengths:** Innovative drone tech, real-time crop monitoring. **Weaknesses:** High initial investment, limited skilled pilots. **Opportunities:** Expanding agri-tech adoption, subsidies. **Challenges:** Weather dependency, government regulations.	8	4	4
10	**Analyze how strategic planning using vision, mission, goals, and objectives can ensure business success.** ** Solution:** **Vision:** Long-term aspiration (e.g., 'To be a global leader in clean energy'). **Mission:** Defines business purpose ('Providing affordable green power'). **Goals:** Broader targets (market expansion, profit growth). **Objectives:** Quantifiable actions (10% sales increase per year).	8	3	3
11	**Evaluate the advantages and limitations of growth strategies such as mergers, acquisitions, and strategic alliances**. ** Solution:** **Advantages:** 1. Market Expansion 2. Shared R&D and technology 3. Economies of scale **Limitations:** 1. Integration challenges 2. Cultural differences 3. Financial risk **Example:** Tata Motors acquiring Jaguar Land Rover for brand expansion.	8	3	3

12	**Design a business model canvas for a mobile app providing real-time traffic and parking solutions.** **Solution** **Key Partners:** IoT data providers, municipalities. **Key Activities:** App development, data analytics. **Value Proposition:** Saves time and fuel, improves urban mobility. **Customer Segments:** Commuters, logistics companies. **Channels:** Mobile app, digital ads. **Revenue Streams:** Subscriptions, advertisements. **Cost Structure:** Server costs, maintenance, marketing.	8	3	3